

TinyOlap Lab Manual: Exploring OLAP Operations in Python

Instructions

1. Open **Google Colab**
2. Copy and paste each code block into a new cell
3. Run cells sequentially to explore each OLAP concept

Environment Setup and Cube Creation

```
!pip install tinyolap

from tinyolap.database import Database
from tinyolap.slice import Slice
import random

# Create database and dimensions
db = Database("edu_cube")

db.add_dimension("metrics").edit().add_many(["Actual", "Plan",
"Deviation"]).commit()
db.add_dimension("years").edit().add_many(["2022", "2023"]).commit()
db.add_dimension("regions").edit().add_many(["North", "South"]).commit()

# Product hierarchy
editor = db.add_dimension("products").edit()
editor.add_many(["Model X", "Model Y"])
editor.add_many("All Products", ["Model X", "Model Y"])
editor.commit()

# Create cube
cube = db.add_cube("sales", ["metrics", "years", "regions", "products"])

# Populate data
for year in ["2022", "2023"]:
    for region in ["North", "South"]:
        for product in ["Model X", "Model Y"]:
            actual = random.randint(1000, 2000)
            plan = random.randint(800, 1900)
            cube["Actual", year, region, product] = actual
            cube["Plan", year, region, product] = plan
            cube["Deviation", year, region, product] = actual - plan
```

Slicing

A slice selects data along a single dimension like sales in 2023.

```

slice_report = Slice(cube, {
    "title": "Slice: Metrics for 2023",
    "header": [{"dimension": "metrics"}],
    "columns": [{"dimension": "years", "member": "2023"}],
    "rows": [{"dimension": "regions"}, {"dimension": "products"}]
})
print(slice_report.as_console_output())

```

Dicing

Dice extracts a subcube by selecting specific values across multiple dimensions.

```

dice_report = Slice(cube, {
    "title": "Dice: 2022 North Model X",
    "header": [{"dimension": "metrics"}],
    "columns": [{"dimension": "years", "member": "2022"}],
    "rows": [{"dimension": "regions", "member": "North"}, {"dimension":
"products", "member": "Model X"}]
})
print(dice_report.as_console_output())

```

Pivoting

Pivoting changes the layout of rows and columns to gain new perspectives.

```

pivot_report = Slice(cube, {
    "title": "Pivot: Products in Columns",
    "header": [{"dimension": "metrics"}],
    "columns": [{"dimension": "products"}],
    "rows": [{"dimension": "regions"}, {"dimension": "years"}]
})
print(pivot_report.as_console_output())

```

Drill-Up

Roll data up to a parent level e.g., aggregating to "All Products".

```

drill_up = Slice(cube, {
    "title": "Drill-Up: Aggregate - All Products",
    "header": [{"dimension": "metrics"}],
    "columns": [{"dimension": "years"}],
    "rows": [{"dimension": "regions"}, {"dimension": "products", "member":
"All Products"}]
})
print(drill_up.as_console_output())

```

Drill-Down

Zoom into detailed members e.g., "Model X" and "Model Y".

```
drill_down = Slice(cube, {
    "title": "Drill-Down: Product Breakdown",
    "header": [{"dimension": "metrics"}],
    "columns": [{"dimension": "years"}],
    "rows": [{"dimension": "regions"}, {"dimension": "products"}]
})
print(drill_down.as_console_output())
```

Cube Rotation

Swap dimensional axes to explore alternate views.

```
rotate_report = Slice(cube, {
    "title": "Cube Rotation: Regions in Columns",
    "header": [{"dimension": "metrics"}],
    "columns": [{"dimension": "regions"}],
    "rows": [{"dimension": "years"}, {"dimension": "products"}]
})
print(rotate_report.as_console_output())
```

Trend Analysis (Time Series)

Observe changes over time

```
trend_report = Slice(cube, {
    "title": "Time Series: Product Performance",
    "header": [{"dimension": "metrics"}],
    "columns": [{"dimension": "years"}],
    "rows": [{"dimension": "products"}]
})
print(trend_report.as_console_output())
```

Exercises: OLAP Cube Exploration with TinyOlap

Exercise 1: Slice Exploration

- Create a slice that focuses on sales in a specific year (e.g., 2023).
- Keep products and regions in rows, and metrics in the header.
- Try slicing by a different region instead of year.

Exercise 2: Dice Drill

- Extract a subcube that includes selected values across multiple dimensions.
- Choose one region, one product, and two different years.
- Compare the resulting data across chosen metrics.

Exercise 3: Pivot Practice

- Pivot one dimension into columns (e.g., products).
- Swap rows and columns to view different perspectives.
- Try pivoting by region, year, or metric and note what changes.

Exercise 4: Trend Over Time

- Explore how performance or deviation shifts across years for each product.
- Use time as one axis, and compare across products or regions.
- Identify any notable upward or downward trends.

Bonus Exercise: Create Your Own View

- Design a custom slice, pivot, or dice operation that tells a unique story.
- Choose any dimension(s) to include or filter.
- Reflect: What patterns emerge? What comparisons become easier?